Task NO:7. utilizing 'Functions' concepts in python programming.

Aim:- To write the python program using 'Functions' concepts in python programming.

7.1 you are developing a small python script to analyze script to analyze and manipulate a list of student grade for a class project. write a python program that satisfies the above requirements using the built-in functions print(), len(), type(), max(), min(), sorted(), reversed() and range().

Algorithm:

1. Start the program.
2. print a welcome message: outputs a simple greeting
3. Determine and print the number of students: uses len() to find the number of elements in the student_names list.
4. print the type of lists: uses type() to show the type of the student_names and student_grades lists.
5. find and print highest and lowest grades: used. max() and min() to determine the highest and lowest values in student_grades.
6. print sorted list of grades: uses sorted() to sort the grades.
7. print reversed. list of grades: uses reversed() to reverse the sorted list and converts it to a list.
8. Generate and print a range of grade indices: uses range() to create a list a indices from 1 to the number of students.
9. stops.

program:

```
def analyze_student_grades();
  # sample data
  student_names = ["Alice", "Bob", "charlie", "Diana"]
  student_grades = [85,92,78,90]

  # 1. print a welcome message
  print ("Welcome to the Student Grades Analyzer! \n")

  # 2. Determine and print the number of students
  num_students = len(student_names)
  print ("Number of Students:", num_students)
  #3. print the type of the student names list and the grades
  print ("\n Type of student_names list:", type (student_names))
  print ("Type of student_grades list:", type (student_grades))
  #4. Find and print the highest and Lowest grade.
  highest_grade = max (student_grades)
  Lowest_grade = min (student_grades)
  print ("\nHighest grade:", highest_grade)
  print ("Lowest grade:", Lowest_grade)
  #5. print the list of grades sorted in ascending order
  sorted_grades = sorted (student_grades)
  print ("\n Sorted grades:", sorted_grades)
  #6. print the list of grades in reverse order
  reversed_grades = list(reversed(sorted_grades))
  print ("Reversed grades:", reversed_grades)
  #7. Generate and print a range of grade indices from
  1 to the number of students grade_indices = list(range
  (1, num_students + 1))
  print("\n Grade indices from 1 to number of students:",
  grade_indices)

  # Run the analysis
  analyze_student_grades()
```

Output:

Welcome to the Student Grades Analyzer!

Number of students: 4

Type of student_names list: <class 'list'>

Type of students_grades list: <class 'list'>

Highest grade: 92

lowest grade: 78

Sorted grades: [78, 85, 90, 92]

Reversed grades: [92, 90, 85, 78]

Grade indices from 1 to number of students: [1, 2, 3, 4].

Q2 You are tasked with creating a small calculator application to help users perform basic arithmetic operations and greet them with a personalized message, your application should perform the following tasks, addition, subtraction, multiplication, division.

Algorithm:

1. Start the program
2. User input for numbers: The program prompts the user to enter two numbers.
3. User input for operation: The program prompts the user to choose an arithmetic operation (addition, subtraction, multiplication, division).
4. Perform operation: Based on the user's choice, the program performs the chosen arithmetic operation using the defined functions.
5. Display Result: The program displays the result of the operation.
6. Stop.

Program:

```
def add(a,b):
  """Return the sum of two numbers."""
  return a+b
def subtract(a,b):
  """Return the difference between two numbers."""
  return a-b
def multiply(a,b):
  """Return the product of two numbers."""
  return a*b
def divide(a,b):
  """Return the quotient of two numbers. Handles division by zero"""
  if b!=0:
  return a/b
  else:
```

Output:

Arithematic operations:

Sum of 10 and 5: 15

Difference between 10 and 5: 5

product of 10 and 5: 50

quotient of 10 and 5: 2.0

Greeting:

Hello, Alice ! Welcome to the program.

```
    return "Error: Division by zero"
def greet(name):
    """ Return a greeting message for the user."""
    return f"Hello, {name}! Welcome to the program."
def main():
    # Demonstrating the use of user-defined functions
    # Arithmetic operations
    num1 = 10
    num2 = 5
    print("Arithmetic operations:")
    print(f"sum of {num1} and {num2}:", add(num1, num2))
    print(f"Difference between {num1} and {num2}:", subtract(num1, num2))
    print(f"product of {num1} and {num2}:", multiply(num1, num2))
    print(f"quotient of {num1} and {num2}:", divide(num1, num2))
    # Greeting the user
    user_name = "Alice"
    print("\n Greeting:")
    print(greet(user_name))
# Run the main function
if _name_ == "_main_":
    main()
```

VELTECH	
EX No.	4
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	—
TOTAL (20)	15
SIGN WITH DATE	[illegible]

Result:- Thus, the Python program using 'Functions' concepts was successfully executed and the output was verified.